

Project Initialization and Planning Phase

Date	19 July 2026
Team ID	XXXXXX
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution automates the credit card approval process using machine learning to improve speed, accuracy, and consistency. It analyzes applicants' financial and demographic information to predict approval decisions, reducing manual effort and minimizing human errors. The solution is deployed as a Flask web application on IBM Cloud/Render, enabling secure and real-time predictions.

Project Overview	
Objective	Develop an intelligent credit card approval prediction system that uses machine learning algorithms to accurately classify applicants as approved or rejected based on their financial profile.
Scope	The project covers data preprocessing, feature engineering, model training using multiple classification algorithms, model evaluation, Flask web application development, and cloud deployment for real-time predictions.
Problem Statement	
Description	Banks receive thousands of credit card applications every day. Manual verification is time-consuming, inconsistent, and prone to human error, making the approval process inefficient.
Impact	An automated prediction system reduces processing time, improves decision accuracy, identifies high-risk applicants, and supports faster, data-driven decision-making.
Proposed Solution	
Approach	Train multiple machine learning classification models on historical applicant data and deploy the best-performing model through a Flask web application for instant credit card approval prediction.
Key Features	- Machine learning-based credit card approval prediction. - Multiple algorithms: Logistic Regression, Decision Tree, Random Forest, and XGBoost.

	- Real-time prediction using a Flask web application. - Simple and user-friendly web interface.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel i5/Ryzen 5 or above
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	20 GB Available Disk Space
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	NumPy, Pandas, Scikit-learn, XGBoost, Matplotlib, Seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle Credit Card Approval Dataset, CSV